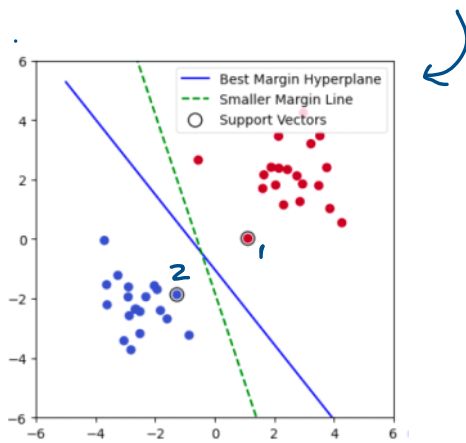
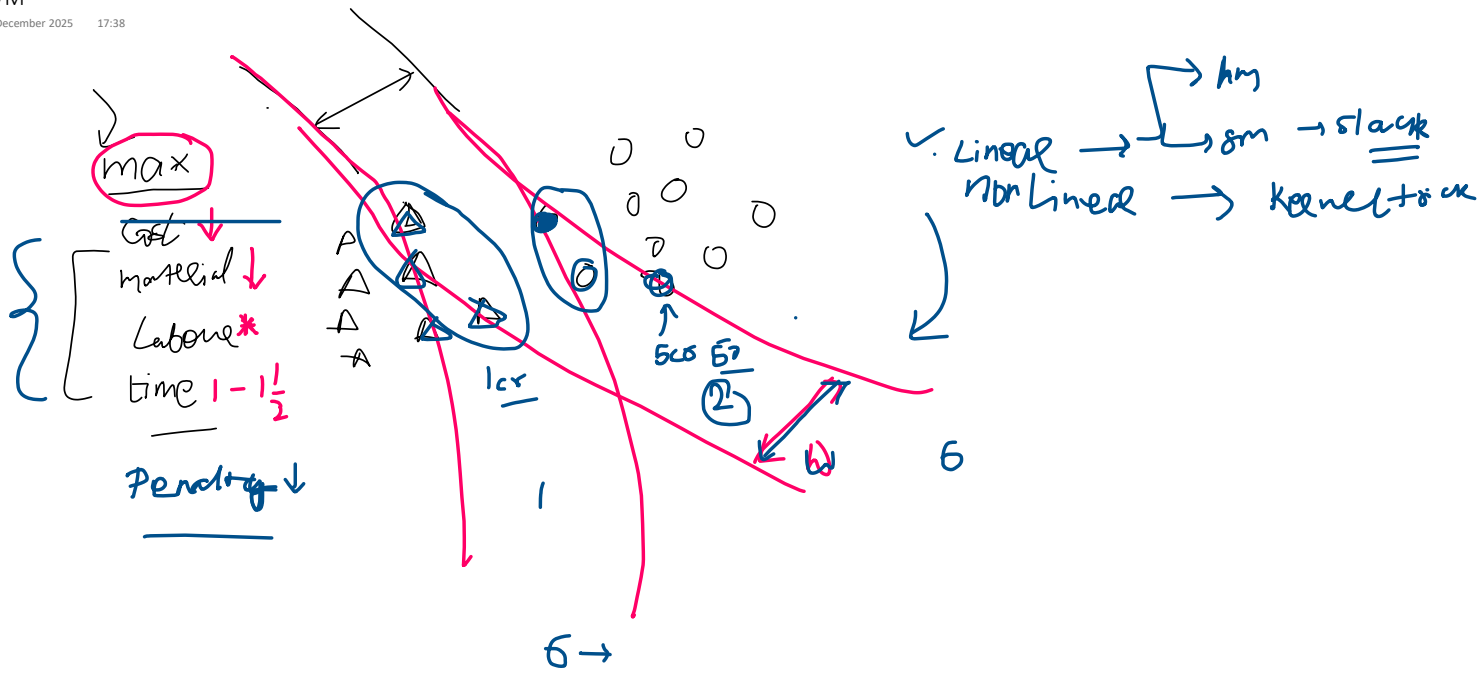
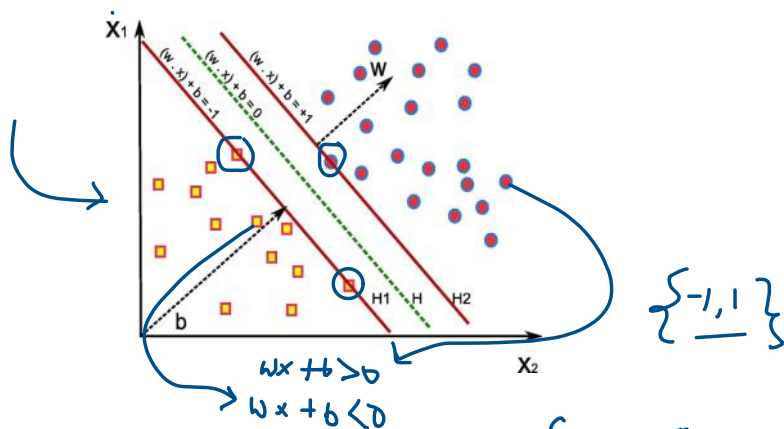


SVM → 1990
2010

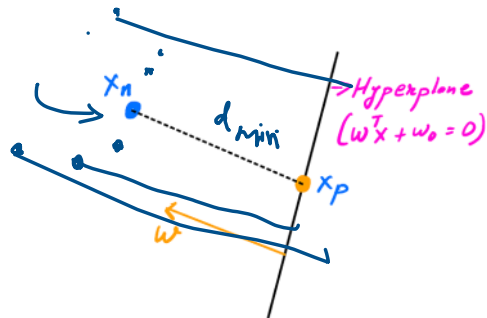


$$\begin{aligned} & \swarrow w x + b > 0 \\ & \searrow w x + b < 0 \end{aligned}$$

$$f(x) = w^T x + w_0$$

$$w^T x = w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$$\text{Class} = \begin{cases} +1 & \text{if } f(x) > 0 \\ -1 & \text{if } f(x) < 0 \end{cases}$$



$$d = \frac{w^T x + w}{\|w\|}$$

$$d = \frac{|w^T x_n + w_0|}{\|w\|}$$

$$\swarrow$$

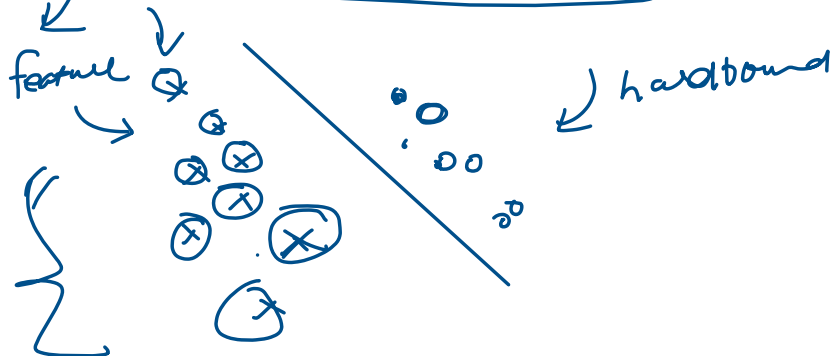
$$\sqrt{w_1^2 + w_2^2}$$

$$y_n = +1 : +(w^T x + w_0)$$

$$y_n = -1 : -(w^T x + w_0)$$

$$\underline{\text{margin}}(x; a, w) = \min_n \frac{y_n (a^T x_n + w a)}{\|w\|}$$

obj fun
linear SVM



$$\min \frac{1}{2} \|w\|^2 \quad \forall y_n (w^T x_n + w_0) > 1 \quad \forall n$$

$$\left[\max_{w, w_0} \left(\min_n \frac{y_n (w^T x_n + w_0)}{\|w\|} \right) \right]$$

$\xi_i \rightarrow$ dummy support

$$\xi_i = 0$$

$$0 < \xi_i < 1$$

$$\xi_i \geq 1 \text{ (misclassified)}$$



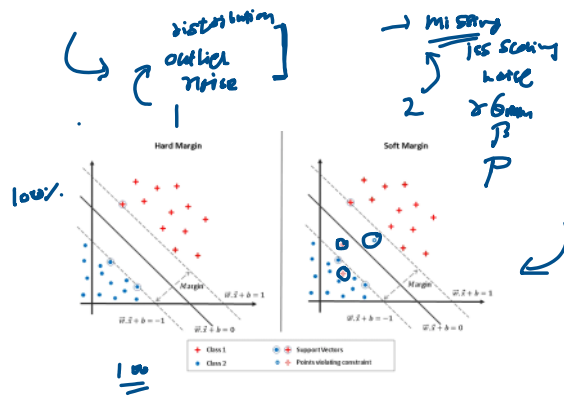
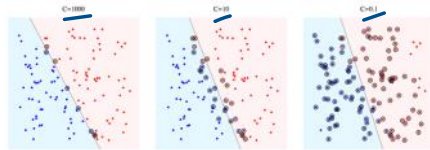
Hyperplane of sup
linear C



TLV

Linear C

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i$$



$$\frac{1}{2} \|w\|^2$$

$$y_i(w x_i + b) \geq 1 - \xi_i$$

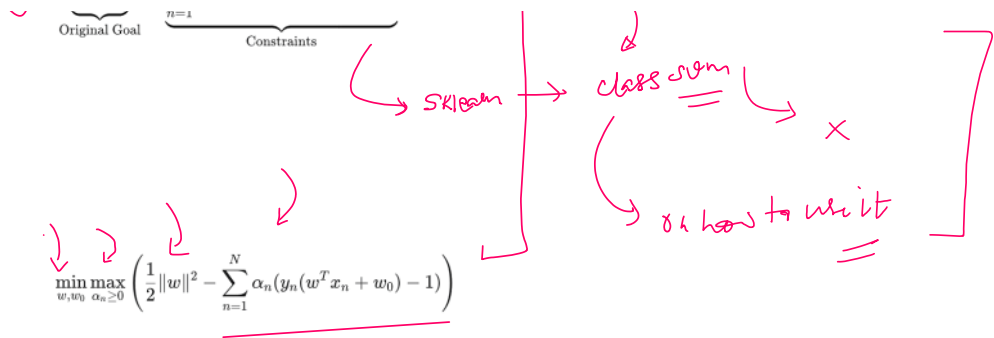
$$\xi_i \geq 0$$

$$\min \frac{1}{2} \|w\|^2 \quad \text{subject to } y_n(w^T x_n + w_0) \geq 1$$

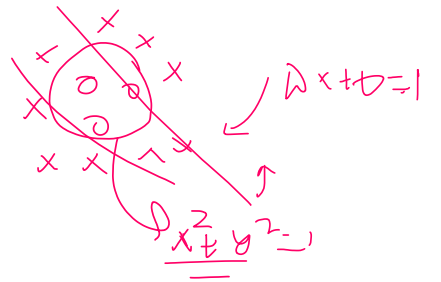
$$\min \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n \quad \text{subject to } y_n(w^T x_n + w_0) \geq 1 - \xi_n, \quad \xi_n \geq 0$$

$$L = \underbrace{\frac{1}{2} \|w\|^2}_{\text{Original Goal}} - \underbrace{\sum_{n=1}^N \alpha_n (y_n(w^T x_n + w_0) - 1)}_{\text{Constraints}}$$

slack → class even



✓ C ✓
 Y → x
 ✓ K ✓

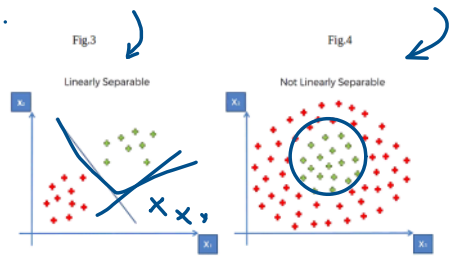


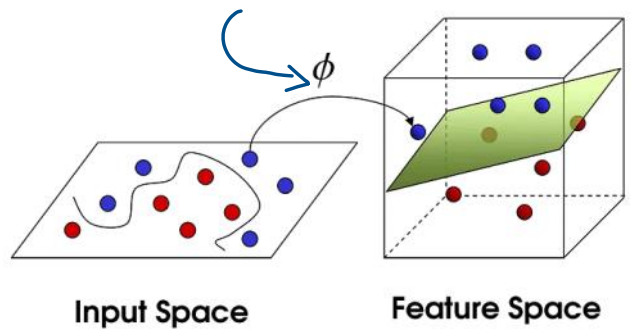
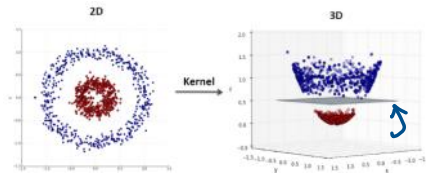
Linear < sm
 hm

Poly $\rightarrow n^3 n^4 n^5 \rightarrow T(0) \rightarrow$

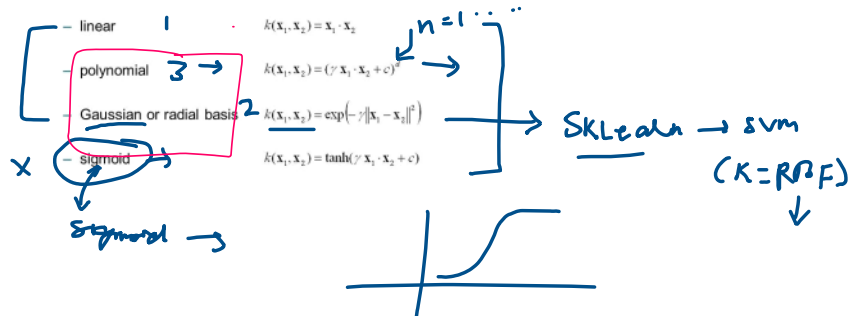
RBF $\rightarrow$

Sigmoid $\rightarrow$





Common kernel functions for SVM



$$\frac{1}{2} \|w\|^2 + y_n (w^T x_n + w_0) \geq 1$$

$$\mathcal{L} = \frac{1}{2} \|w\|^2 + \sum \alpha_n (y_n (w^T x_n + w_0) - 1)$$

$$\min_{w, w_0} \max_{\alpha_n \geq 0} \left(\frac{1}{2} \|w\|^2 - \sum_{n=1}^N \alpha_n (y_n (w^T x_n + w_0) - 1) \right)$$

$$\frac{\partial \mathcal{L}}{\partial w} = w - \sum \alpha_n y_n x_n = 0$$

$$w = \sum \alpha_n y_n x_n$$

δw

$$w = \frac{\sum \alpha_n y_n x_n}{2}$$

$$\frac{\delta L}{\delta w} = -\sum \alpha_n y_n$$

$$\begin{aligned} f(x) &= w^T x + w_0 \\ w &= \sum_{n=1}^N \alpha_n y_n x_n \\ f(x) &= \left(\sum_{n=1}^N \alpha_n y_n x_n \right)^T x + w_0 \\ &= \left(\sum_{n=1}^N \alpha_n y_n (x_n^T x) + w_0 \right) \end{aligned}$$

Linear
polynomial
RBF
Sigmoid